

GUMI, Korea, Republic of

WMO: 471540

Lat: 36.1306N

Lon: 128.3206E

Elev: 50

StdP: 100.72

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -7.3	(d) -21.1	(e) 0.6	(f) -3.3	(g) -19.0	(h) 0.7	(i) -3.0	(j) 6.1	(k) -2.3	(l) 5.4	(m) -1.4	(n) 2.1	(o) 290	(p) 0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.9	(c) 33.8	(d) 25.4	(e) 32.6	(f) 24.8	(g) 31.2	(h) 23.7	(i) 27.4	(j) 30.8	(k) 26.7	(l) 30.3	(m) 25.9	(n) 29.2	(o) 1.7	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.3	28.8	25.7	21.2	28.3	25.0	20.2	27.8	87.4	31.2	84.1	30.5	80.6	29.0	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.4	(b) 4.7	(c) 4.1	(e) -11.4	(f) 35.4	(g) 2.5	(h) 1.0	(i) -13.2	(j) 36.2	(k) -14.7	(l) 36.8	(m) -16.1	(n) 37.3	(o) -17.9	(p) 38.1		
(a) 5.4	(b) 4.7	(c) 4.1	(e) -12.5	(f) 27.6	(g) 2.2	(h) 1.3	(i) -14.1	(j) 28.5	(k) -15.4	(l) 29.3	(m) -16.6	(n) 30.1	(o) -18.1	(p) 31.0		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	-0.4	2.5	7.6	13.3	19.3	23.3	26.2	26.4	21.4	15.2	8.4	1.4
(7)		DBStd	9.84	3.38	3.62	3.53	3.28	2.64	2.07	2.34	2.62	2.55	3.01	3.92	3.43
(8)		HDD10.0	984	323	212	93	9	0	0	0	0	0	2	77	267
(9)		HDD18.3	2464	582	444	332	151	20	1	0	0	4	106	299	526
(10)		CDD10.0	2362	0	2	19	110	288	399	503	509	341	163	28	0
(11)		CDD18.3	801	0	0	0	2	51	149	244	251	96	8	0	0
(12)		CDH23.3	7094	0	0	1	62	535	1171	2322	2466	519	18	0	0
(13)		CDH26.7	2441	0	0	0	6	131	346	872	995	91	0	0	0
(14)	Wind	WSAvg	1.5	1.8	1.7	1.9	1.8	1.6	1.4	1.2	1.2	1.1	1.1	1.4	1.6
(15)	Precipitation	PrecAvg	1093	18	28	47	76	83	124	235	244	142	43	35	19
(16)		PrecMax	1744	59	89	113	164	205	351	641	599	439	137	132	47
(17)		PrecMin	638	1	1	4	15	21	11	96	52	10	1	0	0
(18)		PrecStd	253	13	23	27	40	42	76	123	132	108	37	31	14
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	10.2	17.0	22.3	27.4	32.1	32.8	34.8	35.7	31.0	25.2	21.1	12.3	
(20)		0.4% MCWB	4.7	12.6	12.8	13.6	18.2	20.2	26.6	26.2	23.6	19.0	15.4	9.2	
(21)		2% DB	8.1	13.3	19.6	25.1	29.6	31.1	33.5	34.3	28.8	23.6	18.1	10.1	
(22)		2% MCWB	3.7	7.6	10.4	13.8	16.9	20.0	25.9	25.9	22.0	17.9	13.5	5.9	
(23)		5% DB	6.4	10.8	17.4	23.1	28.0	30.0	32.3	33.1	27.5	22.1	16.4	8.5	
(24)		5% MCWB	2.6	5.4	9.3	12.6	16.4	20.1	25.1	25.6	21.4	16.4	12.5	5.3	
(25)		10% DB	5.0	8.7	14.9	20.8	26.5	28.8	31.1	31.6	26.3	20.7	14.7	7.0	
(26)		10% MCWB	1.4	3.9	8.8	11.7	16.1	19.8	24.7	25.2	20.9	16.0	11.0	4.1	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	6.6	14.3	14.8	17.4	23.4	24.7	28.3	28.1	25.9	22.5	16.8	10.7	
(28)		0.4% MCDB	9.1	16.3	18.3	22.6	24.7	27.1	30.9	31.3	27.3	23.7	19.4	12.0	
(29)		2% WB	4.7	8.8	12.9	15.7	19.8	23.3	27.6	27.4	24.5	20.5	14.8	7.5	
(30)		2% MCDB	6.8	10.6	15.7	20.3	24.6	26.4	30.5	31.3	27.1	21.8	17.2	8.9	
(31)		5% WB	3.3	6.1	11.3	14.6	18.6	22.4	26.9	26.7	23.5	18.5	13.3	5.7	
(32)		5% MCDB	5.6	9.8	17.0	19.5	24.1	26.1	30.3	30.9	25.8	20.1	15.4	8.0	
(33)		10% WB	2.0	4.5	9.0	13.2	17.6	21.5	26.2	26.0	22.3	16.7	12.0	4.2	
(34)		10% MCDB	4.5	7.9	13.7	18.6	23.3	25.8	29.1	29.8	24.7	19.6	14.0	6.8	
(35)	Mean Daily Temperature Range	MDBR	8.8	9.5	11.1	11.9	12.2	9.5	7.5	7.9	8.9	10.4	9.4	8.3	
(36)		5% DB MCDBR	10.4	12.6	15.3	16.7	15.4	11.9	9.5	10.2	10.4	11.9	11.3	10.3	
(37)		5% DB MCWBR	6.8	7.8	7.8	6.9	5.2	3.3	2.9	3.1	4.2	5.5	6.4	7.0	
(38)		5% WB MCDBR	9.4	11.2	12.6	12.0	11.3	7.9	8.0	8.8	7.5	7.2	9.1	9.0	
(39)		5% WB MCWBR	6.9	8.2	7.8	6.6	5.3	3.1	3.4	3.5	4.1	4.6	6.0	6.8	
(40)	Clear-Sky Solar Irradiance	taub	0.405	0.454	0.536	0.570	0.580	0.637	0.593	0.549	0.480	0.455	0.430	0.399	
(41)		taud	2.039	1.950	1.786	1.761	1.780	1.723	1.892	2.002	2.141	2.083	2.065	2.075	
(42)		Ebn at Noon	749	758	730	734	734	692	719	743	776	751	717	725	
(43)		Edh at Noon	129	159	205	222	221	233	196	172	143	138	126	118	
(44)	All-Sky Solar Radiation	RadAvg	2.67	3.38	4.36	4.99	5.59	5.08	4.37	4.40	3.93	3.61	2.66	2.39	
(45)		RadStd	0.17	0.32	0.42	0.38	0.51	0.40	0.59	0.58	0.44	0.31	0.32	0.16	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+72	N/A

Nomenclature: See separate page